5. Exploratory Data Analysis and geospatial analysis

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2. Summary Statistics
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  - Detect Inconsistencies (eg: speed 0 but distance increasing)

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    $df.groupby("Hour")["Speed"].mean()$
  - Vehicle stat
    $df.groupby("VehicleID")["Speed"].agg(["mean","max","min","std"])$

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  - Interpretation: Traffic is skewed (mean ¿ median), High variability (unstable traffic conditions), Likely congestion in some periods
- Correlation Between Variables
  $df[["Speed","TravelTime","Distance"]].corr()$

Data Visualization

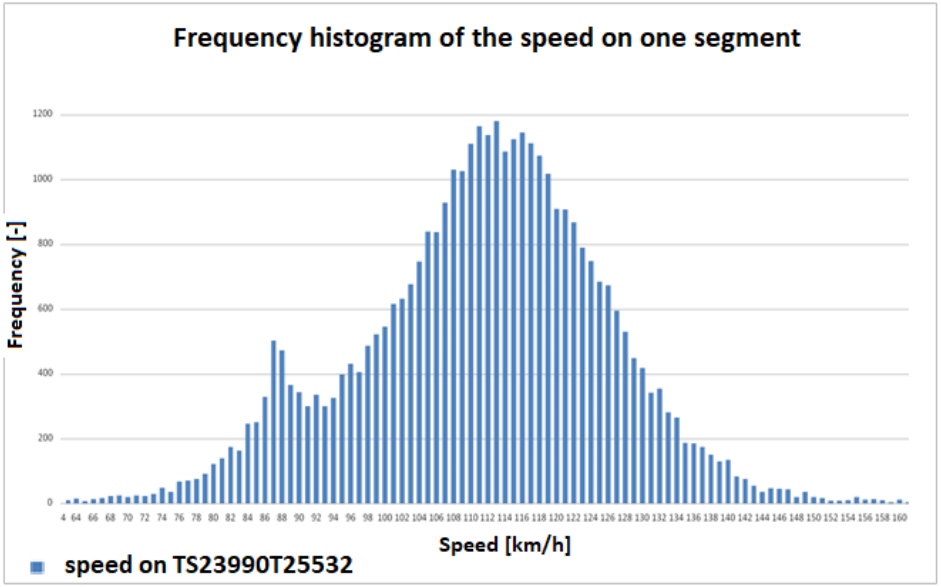
Visualization = turning numbers into insight.
Common libraries in Python:

- Matplotlib → basic plots
- Seaborn → advanced & beautiful plots
- Plotly → interactive charts
- Folium → maps (great for GPS data)

Data Visualization

- Histogram

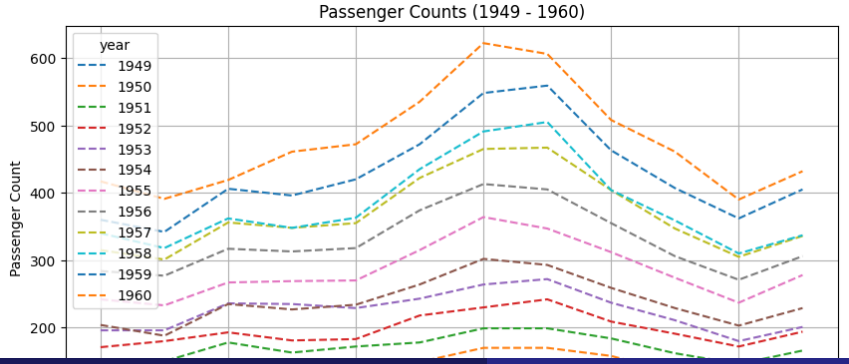
```
import matplotlib.pyplot as plt

plt.hist(df["Speed"], bins=20)
plt.title("Speed Distribution")
plt.xlabel("Speed (km/h)")
plt.ylabel("Frequency")
plt.show()
```



Data Visualization

- Line Plot: used to show trends in data by connecting data points with straight lines

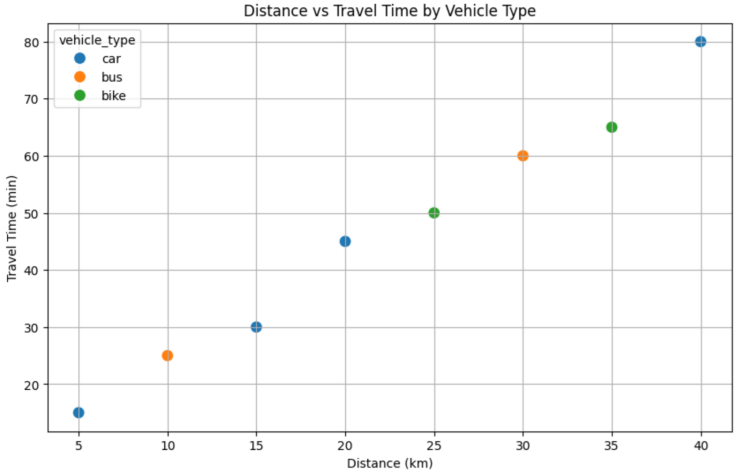
```
import matplotlib.pyplot as plt
import pandas as pd

# 1. Sample Transportation Data (Hours vs Passengers)
data = {
    'Hour': [6, 7, 8, 9, 10, 11, 12, 13, 14, 15],
    'Passengers': [50, 150, 300, 200, 100, 80, 120, 140, 110, 160]
}
df = pd.DataFrame(data)

# 2. Create the line plot
plt.figure(figsize=(10, 5))
plt.plot(df['Hour'], df['Passengers'], color='blue', marker='o', linestyle='-',

# 3. Add Labels and Title
plt.title('Hourly Public Transport Ridership', fontsize=14)
plt.xlabel('Hour of Day', fontsize=12)
plt.ylabel('Number of Passengers', fontsize=12)
plt.grid(True)

# 4. Display the plot
plt.show()
```



Data Visualization

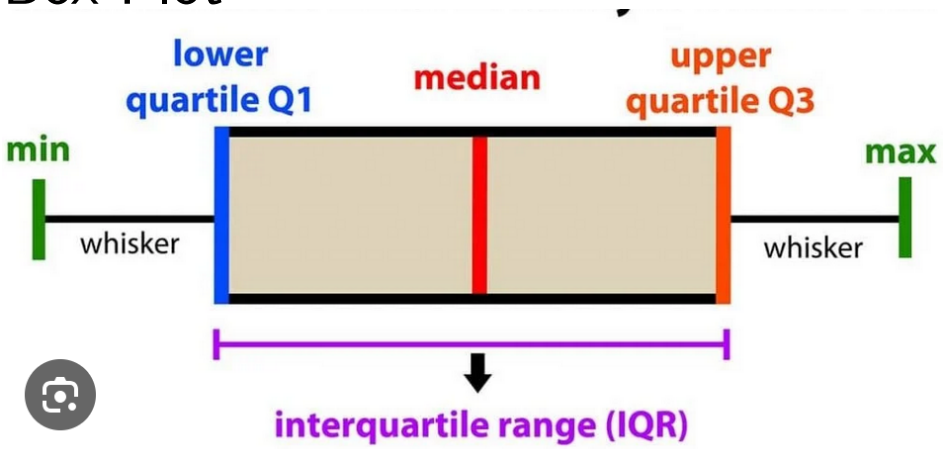
- Scatter Plot: Eg: Relationship between distance and travel time

```python
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# Sample Transportation Data
data = {
    'distance': [5, 10, 15, 20, 25, 30, 35, 40],
    'travel_time': [15, 25, 30, 45, 50, 60, 65, 80],
    'vehicle_type': ['car', 'bus', 'car', 'car', 'bike', 'bus', 'bike', 'car']
}
df = pd.DataFrame(data)

# Create Scatter Plot
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='distance', y='travel_time', hue='vehicle_type', s=100

# Add Labels and Title
plt.title('Distance vs Travel Time by Vehicle Type')
plt.xlabel('Distance (km)')
plt.ylabel('Travel Time (min)')
plt.grid(True)

# Show Plot
plt.show()
```



Data Visualization

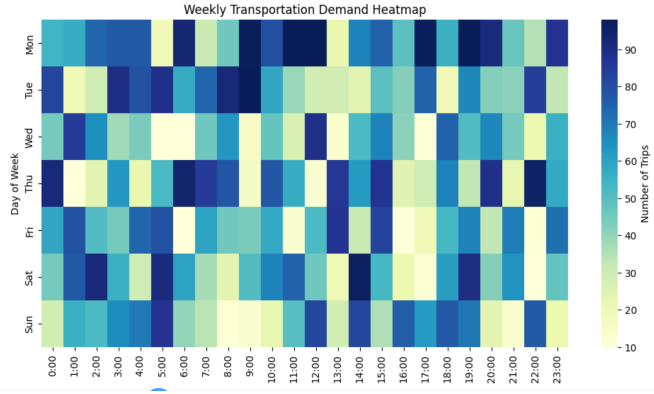
- Box Plot



Data Visualization

- Heat map

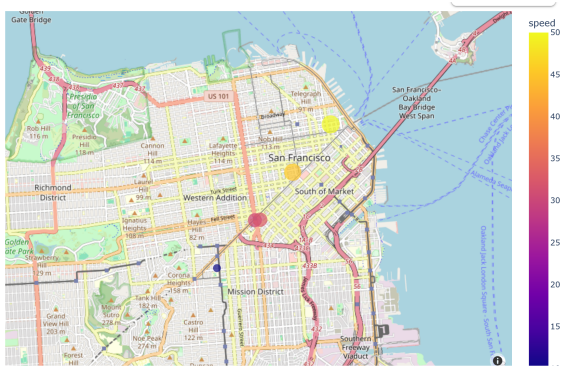
```python
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

# 1. Create synthetic transport data (Hours vs Days)
np.random.seed(0)
# Data: 7 days (Rows), 24 hours (Cols) - values = number of trips
data = np.random.randint(10, 100, size=(7, 24))
days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
hours = [f"{i}:00" for i in range(24)]
df = pd.DataFrame(data, index=days, columns=hours)

# 2. Generate Heatmap
plt.figure(figsize=(12, 6))
sns.heatmap(df, cmap="YlGnBu", annot=False, cbar_kws={'label': 'Number of Trips'})
plt.title("Weekly Transportation Demand Heatmap")
plt.xlabel("Hour of Day")
plt.ylabel("Day of Week")
plt.show()
```



Data Visualization

- GPS scattered map

```python
import plotly.express as px
import pandas as pd

# 1. Create synthetic transportation data
data = {
    'vehicle_id': ['Bus_A', 'Bus_B', 'Taxi_1', 'Taxi_2', 'Bus_A'],
    'lat': [37.7749, 37.7849, 37.7649, 37.7949, 37.7750],
    'lon': [-122.4194, -122.4094, -122.4294, -122.3994, -122.4180],
    'speed': [30, 45, 10, 50, 32]
}
df = pd.DataFrame(data)

# 2. Create the Scatter Mapbox plot
fig = px.scatter_mapbox(
    df,
    lat="lat",
    lon="lon",
    color="speed",
    size="speed",
    hover_name="vehicle_id",
    zoom=12,
    height=600,
    title="Real-time Transportation GPS Data"
)

# 3. Configure the map style
fig.update_layout(mapbox_style="open-street-map")
fig.update_layout(margin={"r":0,"t":40,"l":0,"b":0})

# 4. Display the map
fig.show()
```



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  - Shapely $\rightarrow$ geometric operations
  - Folium $\rightarrow$ interactive maps
  - Geopy $\rightarrow$ location services

Geospatial Analysis

- Plot GPS

```python
import pandas as pd
import plotly.express as px

# 1. Create a sample DataFrame with GPS points
# In a real scenario, you would load data from a CSV or other source
data = {
    'Latitude': [8.9779, 9.0183, 9.0435],
    'Longitude': [38.7981, 38.7694, 38.7610],
    'Location': ['Bole Airport', 'Meskel Square', 'AAU']
}



df = pd.DataFrame(data)

# 2. Create an interactive scatter mapbox plot
fig = px.scatter_mapbox(df,
                        lat="Latitude",
                        lon="Longitude",
                        hover_name="Location", # Display location name on hover
                        zoom=3,                # Set initial zoom level
                        height=400,
                        width=600)

# 3. Update map style to OpenStreetMap (requires internet connection for tiles)
fig.update_layout(mapbox_style="open-street-map")
fig.update_layout(margin={"r":0,"t":0,"l":0,"b":0})

# 4. Display the map
# In a Python script, this opens in your default web browser
fig.show()
```

Discussion

Hands on experiance (Lab work)